

Program : Diploma in Civil & Environmental Engineering	
Course Code : 3352	Course Title: Mechanics of Fluids
Semester : 3	Credits: 3
Course Category: Programme Core Course	
Periods per week: 3 (L:2 T:1 P:0)	Periods per semester: 45

Course Objectives:

- To determine static pressure on immersed vertical surfaces.
- To explain the flow through pipes and open-channels.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic knowledge of Mathematics	2002	Mathematics II	2
Basic knowledge of Mechanics	2021	Engineering Mechanics	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Determine Hydrostatic Pressure on Trapezoidal Gravity Dams with water-face vertical.	10	Applying
CO2	Explain concepts of Buoyancy and Bernoulli's Theorem	11	Applying
CO3	Articulate the theory of losses in pipe flow and uniform flow in open-channels	11	Applying
CO4	Explain the concepts of Non-uniform Flows in Open-channel	11	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-
CO4	3	-	-	-	-	-	-

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Determine Hydrostatic Pressure on Trapezoidal Gravity Dams with water-face vertical.		
M1.01	Explain different fluids and its properties	2	Understanding
M1.02	Solve pressure measurement problems in piezometers and U-tube manometers	4	Applying
M1.03	Determine the pressure force on submerged vertical surfaces	4	Applying
Contents: Introduction to the subject-Fluid properties such as mass density, specific weight, viscosity, specific gravity and surface tension (no derivations required); Classification of Fluids – Newtonian and Non-Newtonian Fluids. Fluid statics- Hydrostatic Law and its application (no derivations required); Atmospheric Pressure, Gauge Pressure, Vacuum Pressure, Absolute Pressure. Measurement of fluid pressure using piezometers and U-tube manometers only, Simple Numerical Problems. Pressure Force on immersed vertical surface (no derivations required). Practical application of total pressure on trapezoidal gravity dams with water-face vertical, Only simple numerical problems.			
CO2	Explain concepts of Buoyancy and Bernoulli's Theorem		
M2.01	Explain Buoyancy and stability of floating bodies	2	Understanding
M2.02	Determine flow rate through pipes	4	Applying

M2.03	Compute discharge through a sharp-edged circular orifice	4	Applying
	Series Test -I	1	
Contents: Buoyancy and Floatation: Buoyant force, Principle of floatation, stability of floating and submerged bodies, metacentre and metacentric height (Only discussion of theory. No derivations or numerical problems required.) Forces considered in describing fluid motion. Euler's equation and Bernoulli's equation – assumptions involved and significance of its terms (No derivation required). Applications of Bernoulli's equation- Venturimeter, Orificemeter and Pitot tube. Only simple numerical problems. Hydraulic coefficients of orifices (No derivation required). and their experimental determination, Discharge through a sharp-edged circular orifice (No derivation required). Only simple numerical problems.			
CO3	Articulate the theory of losses in pipe flow and uniform flow in open-channels		
M3.01	Describe losses in pipe flow	4	Applying
M3.02	Compute discharge through rectangular channel for uniform flow	5	Applying
M3.03	Explain Flow measurement in channels	2	Understanding
Contents: Pipe flow- Equations for determination of major and minor losses in pipes (No derivation required), Only simple numerical problems. Hydraulic gradient line and Total energy line, Flow through pipes in series and parallel (No derivation required). Open channel flow – comparison between pipe flow and open channel flow, velocity distribution in open channels, types of channels, type of flow, Reynold's Number and Froude's Number, Geometric elements of channel section, uniform flow computations using Chezy's equation and Manning's formula for rectangular channels (No derivations required); Only simple numerical problems. Flow measurement in channels – notches and weirs. Discharge equations of rectangular weir, triangular weir and trapezoidal weir (No derivation required).			
CO4	Explain the concepts of Non-uniform Flows in Open-channel		
M4.01	Explain the concept of Specific Energy	2	Understanding
M4.02	Sketch the water surface profiles in Gradually varied flows through rectangular prismatic channels	4	Understanding
M4.03	Explain the concept of Hydraulic Jump	4	Applying

	Series Test – II	1	
Contents: Specific energy- its equation, variation with depth and critical flow (No derivation required). Gradually varied flow- Dynamic equation of gradually varied flow (No derivation required). Types of water surface profiles in rectangular prismatic channels (only sketching of profiles). Rapidly varied flow-Hydraulic jump-conjugate or sequent depths, expression for sequent depths and energy loss for a hydraulic jump in horizontal rectangular channels (No derivation required); Only simple numerical problems.Types of hydraulic jump and its application.			

Text / Reference:

T/R	Book Title/Author
T1	Modi P. N. and S. M. Seth, Hydraulics & Fluid Mechanics, S.B.H Publishers, New Delhi, 2002.
T2	Subramanya K., Theory and Applications of Fluid Mechanics, Tata McGraw-Hill, 1993.
R1	Streeter.V.L. Fluid Mechanics, Mc Graw Hill Publishers.
R2	Frank.M.White, Fluid Mechanics, Mc Graw Hill, 2013.

Online resources:

Sl.No.	Website Link
1	https://www.nptel.ac.in/